

# Gurleen Kaur

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## EDUCATION

### New York University, Tandon School of Engineering

Master of Science in Computer Science — GPA: 3.67/4.0

Brooklyn, NY

Sep 2025 – May 2027

### Thapar University

Bachelor of Technology in Computer Engineering — CGPA: 9.1/10.0

Patiala, India

Aug 2019 – Aug 2023

## TECHNICAL SKILLS

**Languages:** Java, Python, C/C++, SQL, JavaScript, HTML, CSS

**Frameworks & Tools:** Spring Boot, Dropwizard, Django, FastAPI, GraphQL, Flutter, Docker, Kubernetes, Git, Travis CI/CD

**Cloud & Databases:** AWS (DynamoDB, SQS, ECS), PostgreSQL, Firebase, Redis, FAISS (Vector DB)

**Messaging & AI/ML:** Apache Kafka, RAG Pipelines, LLMs, Sentence Transformers, TFLite, Prompt Engineering

## WORK EXPERIENCE

### Flipkart (under Walmart)

Software Development Engineer I

Bengaluru, India

Jun 2024 – Aug 2025

- Designed and developed the **Price Lock** backend system using **Java, Dropwizard, and Guice**, serving **500M+** users with low-latency microservices and RESTful APIs.
- Led **code reviews** and enforced engineering best practices; collaborated with cross-functional teams in an **Agile** environment to deliver features on schedule.
- Executed **end-to-end testing** (unit, integration, regression) ensuring scalability, fault tolerance, and production reliability.

### Synopsys India Pvt. Ltd.

R&D Engineer I

Noida, India

Jul 2023 – Jun 2024

- Engineered **Power Lint** static analysis rules in **C++** within a large-scale EDA architecture, reducing tool response time by **40%** and improving user satisfaction by **20%**.
- Resolved complex architectural defects in production systems used by **Apple, Qualcomm, and Sony**, directly impacting chip design workflows.
- Collaborated with globally distributed teams applying strong **debugging, root cause analysis, and cross-team communication** in a fast-paced R&D environment.

### Amazon

Software Development Engineer Intern

Hyderabad, India

Jan 2023 – Jun 2023

- Implemented **DynamoDB ACID transactions** to consolidate redundant database operations, reducing infrastructure costs by **\$3,400/month** while improving data consistency.
- Architected a **reusable document generation service** using AWS, adopted across multiple internal teams to standardize workflows.
- Improved system reliability through **comprehensive unit and integration testing**, code reviews, and system design discussions.

## PROJECTS

**InsightFlow** | Python, FastAPI, FAISS, Google Gemini, PostgreSQL, Sentence Transformers | GitHub

- Built a **Text-to-SQL RAG pipeline** enabling natural language queries over **100+ database tables** using **FAISS vector search** and Sentence Transformers for semantic schema retrieval.
- Reduced LLM token costs by optimizing context injection; designed a modular **FastAPI** backend with extensible data source connectors and prompt engineering strategies.

**NOMZ** | Django, React, AWS (EC2, S3), PostgreSQL, Travis CI/CD | Live

- Built a **full-stack restaurant recommendation platform** with Django REST backend and React frontend, using a **collaborative filtering engine** and NYC DOHMH open data for personalized suggestions.
- Implemented **real-time chat**, restaurant claim portal; deployed on **AWS** with **Travis CI/CD** for automated testing and zero-downtime releases.

**FunFit** | Django, Flutter, GraphQL, Google ML Kit, TFLite, Firebase | GitHub | Devpost

- Developed a **cross-platform fitness app** with **on-device ML** (TFLite, Google ML Kit) for real-time exercise detection and a gamification system rewarding users based on food nutrition via computer vision.
- Implemented a **GraphQL API** backend with Django for efficient, flexible data fetching across the Flutter mobile client.

## LEADERSHIP & AWARDS

**Exam Proctor**, NYU CS Dept (Fall 2025): Supervised **100+ students**, ensuring exam integrity. **Executive Member**, Microsoft Learn Student Chapter: Organized hackathons and technical workshops, mentoring junior developers. **Winner**, Hack4She 2021 | **Best Beginner**, HooHacks — Recognized for outstanding problem-solving.